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BECAMEX BUSINESS SCHOOL



GROUP PROJECT

Mega Hub Location and Logistics Strategy for Mekong KVD

Course: SCLM 429 - Transportation And Logistics Managment

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1. Introduction

1.1. Company Background

Mekong KVD is a fast-growing mid-sized Vietnamese FMCG manufacturer that specialises in everyday pantry staples – basic fish sauce, soy sauce and cooking oils. The company operates on a Business-to-Business (B2B) wholesale model, delivering heavy-freight pallets to 15 major supermarket and distribution partners. Key partners are highly concentrated in Western HCMC and the Mekong Delta region, with high volume destinations such as Bách Hóa Xanh Regional DC in Vĩnh Long, GO! Cần Thơ, MM Mega Market Hưng Lợi, etc.

1.2. Current operation challenges

The market is growing but Mekong KVD's supply chain is constrained by 4 fragmented & outdated facilities in western HCMC. The company suffers from severe included:

Logistical Inefficiency: The geographical distance between the warehouses leads to considerable issues in inventory management and distribution planning.

High fleet maintenance costs: Routes and locations are not optimized and thus high maintenance and transportation costs are incurred.

High product breakage rates: As the company uses mainly glass components for its products, the movement of goods between stages of the supply chain can cause avoidable damage like breakage.

Mekong KVD thus took a strategic decision to consolidate its operations into one "Mega Hub" to address these pressing issues.

1.3. Solution & Report Purpose

- **Strategic solution:**Mekong KDV could consolidate its four existing facilities into a single new mega-hub. This hub would integrate all the functions of the previous sites such as processing, bottling, and B2B operations, etc into one location.

- **Specific objectives of the report:** Research and Comparison of 3 Possible Industrial Parks in Selected Region. Final selection will be based on Weighted Scoring Model under strict constraints:

+ Strict CapEx budget ceiling of VND 300 Billion (~$12 Million USD) for land purchase and initial construction (making premium IPs in HCMC, Binh Duong and most of Dong Nai

out of budget).

+ A strong preference for IP's with 30+ years remaining on their government master leases to maximize Return on Investment (ROI).

+ Geographical location: Ensuring cost-effectiveness and transportation connections between the company's 15 branches in the western region and Ho Chi Minh City.

- Transportation & Material Handling Strategy: Cost- effectiveness comparison of standard road transport methods versus inland waterway & barge transport in the Mekong Delta to major distribution hubs to evaluate the optimal transport mode.

Furthermore, such shipping can help to reduce breakage of glass bottles during transport.

Ultimately, this report can aid the Mekong Delta region in building a more cost-effective logistics roadmap, reducing operating costs, improving product safety and minimizing breakage, and supporting the company's long term growth in Vietnam's FMCG market.

2. General Area Selection

2.1. Objective

The purpose of this section is to identify the most appropriate geographical area for the introduction of Mekong KVD's new Mega Hub. Selecting a suitable location is the most important strategy in managing a supply chain network because it directly affects transportation costs, delivery lead time, operational efficiency, and customer service performance.

Currently, Mekong KVD operates four fragmented facilities located in western Ho Chi Minh City. This decentralized network resulted in insufficient logistic operations and significant product breakage during transportation. To overcome this problem, the company intends to consolidate its production, bottling, warehousing, and outbound logistics activities into a single integrated Mega Hub.

So it is essential to determine the most appropriate geographical region before choosing a specific industrial park. Through the Center of Gravity Method (COG), we can analyze a location that minimizes the overall weighted transportation distance between the Mega Hub and the company's major B2B customers.

2.2. Customer Distribution Analysis

Mekong KVD applied a business-to-business (B2B) wholesale model to distribute its products. Based on the available data, the company's partners are mostly located in the Mekong Delta region or in western Ho Chi Minh City. Because each region has a different customer scale and needs, the number of weekly deliveries range from 80 to 250 pallets. Consequently, transportation demand is uneven, which impacts the determination of optimal facility locations.

To better understand the logistics network of this company, all partners' locations are mapped according to their geographical coordinates and weekly pallet demand. To be more specific, figure 1 demonstrates the spatial distribution of key customers of Mekong KVD, whereas Table 1 summarizes the delivery volume of each destination.

In addition, there are some locations that have a high demand, including Long An, Tien Giang, Vinh Long, Dong Thap, and Can Tho. Consequently, the Mega Hub is likely to be located near these areas of high demand to minimize transportation distances, while still ensuring efficient product delivery to the remaining customers.

Partners	Location	Weekly Pallets
Bách Hóa Xanh Regional DC (Bình Minh)	Vĩnh Long	250
GO! Cần Thơ	Cần Thơ	200
MM Mega Market Hưng Lợi	Cần Thơ	180
Co.opmart Vĩnh Long	Vĩnh Long	160
Co.opmart Tân An	Tây Ninh	150
Co.opmart Mỹ Tho	Đồng Tháp	140
Co.opmart Long Xuyên	An Giang	140
Co.opmart Cao Lãnh	Đồng Tháp	130
AEON Mall Bình Tân	Bình Tân, HCMC	120
GO! Bến Tre	Bến Tre, Vĩnh Long	120
Co.opmart Sa Đéc	Đồng Tháp	110
Co.opmart Rạch Giá	An Giang	100
Co.opmart Châu Đốc	An Giang	90
Co.opmart Gò Công	Đồng Tháp	90
Satra Củ Chi	Củ Chi, HCMC	80

Table 1. Customer Locations and Weekly Demand

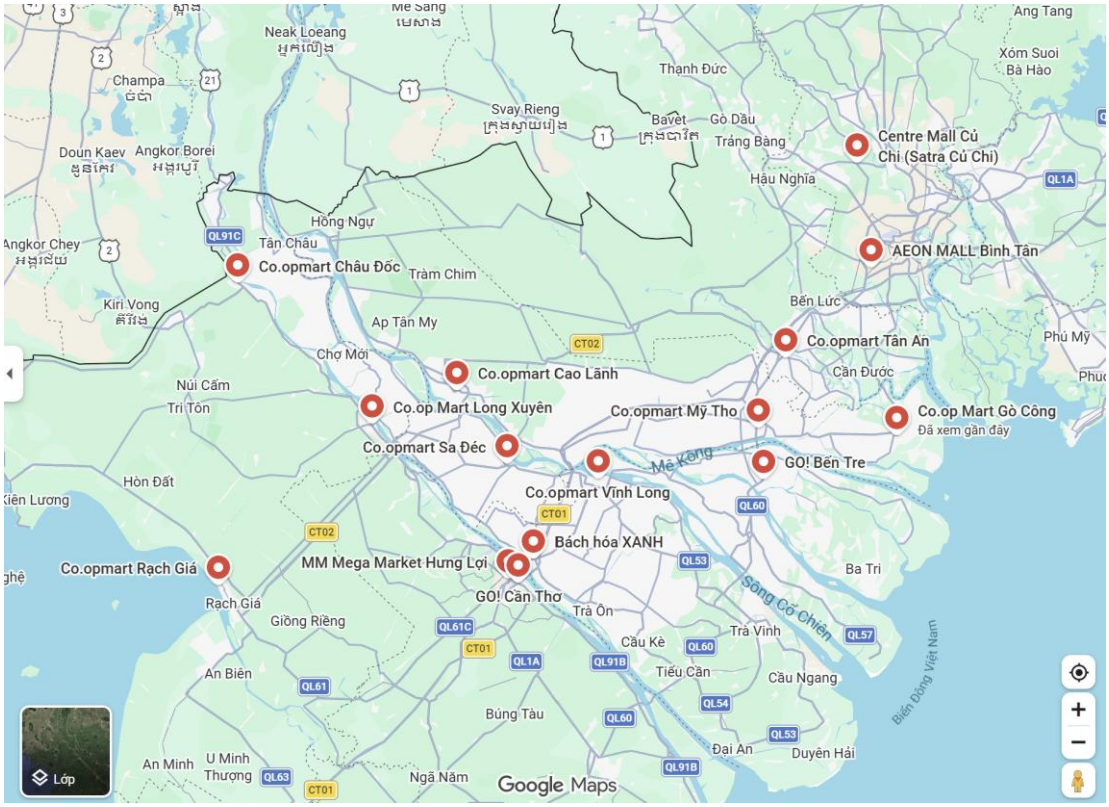


Figure 1. Distribution of Major B2B Customers

2.3. Methodology: Center of Gravity Analysis

To find the best appropriate location for the construction of the Mekong KVD's Mega Hub, the team applied the Center of Gravity (COG) method. This quantitative technique has a variety of applications in supply chains as a site selection support method that will help to identify a geographic point that minimizes the total weighted transportation distance.

This method can calculate a weighted average location by considering both customer locations and shipment volumes. It is more convincing and certain to make a decision based on intuition. Customers with higher weekly pallet requirements exert a significant influence on the final location decision.

The weighted coordinates are calculated based on the following formulas:

$$X = \frac{\sum(x_i \times w_i)}{\sum w_i}$$

$$Y = \frac{\sum(y_i \times w_i)}{\sum w_i}$$

where:

- X= weighted longitude
- Y= weighted latitude
- x_i= longitude of customer *i*
- y_i= latitude of customer *i*
- w_i= weekly pallet demand of customer *i*

2.4. Data Collection

Center of Gravity analysis requires three types of data:

- Customer locations (latitude and longitude)
- Weekly pallet demand
- Customer identification and destination details

Data on weekly pallet demand is provided in the project summary, while geographic coordinates are obtained via Google. Then, this data is collected and calculated in Microsoft Excel to perform the calculations.

Before implementing this calculation, the coordinates are compiled and recorded in decimal format with four decimal places to ensure consistency throughout the analysis process.

B2B Customer	Location	Latitude (X)	Longitude (Y)	Weekly Pallets
Bách Hóa Xanh Regional DC (Bình Minh)	Vĩnh Long	105,8209	10,0706	250
GO! Cần Thơ	Cần Thơ	105,7844	10,0151	200
MM Mega Market Hưng Lợi	Cần Thơ	105,7609	10,0235	180
Co.opmart Vĩnh Long	Vĩnh Long	105,971	10,2551	160
Co.opmart Tân An	Tây Ninh	106,4112	10,5351	150
Co.opmart Mỹ Tho	Đồng Tháp	106,3475	10,3733	140
Co.opmart Long Xuyên	An Giang	105,4405	10,3819	140
Co.opmart Cao Lãnh	Đồng Tháp	105,6413	10,4598	130
AEON Mall Bình Tân	Bình Tân, HCMC	106,6119	10,7429	120

GO! Bến Tre	Bến Tre, Vĩnh Long	106,3599	10,2544	120
Co.opmart Sa Đéc	Đồng Tháp	105,7583	10,2902	110
Co.opmart Rạch Giá	An Giang	105,1188	9,9608	100
Co.opmart Châu Đốc	An Giang	105,1245	10,7073	90
Co.opmart Gò Công	Đồng Tháp	106,6731	10,3549	90
Satra Củ Chi	Củ Chi, HCMC	106,5803	10,9839	80

Table 2. Latitude, Longitude, and Weekly Pallet Demand

2.5. Center of Gravity Calculation

The calculations were performed using Microsoft Excel based on the equations presented in Section 2.3. For each customer location, the latitude and longitude values are multiplied by the corresponding weekly pallet demand. Subsequently, these values are summed and divided by the total weekly pallet volume.

The calculation procedure is summarized below:

Step 1: Calculate X-coordinates

$$X = \frac{(105{,}8209\ x\ 250) + (105{,}7844\ x\ 200) + \ldots + (106{,}0281\ x\ 90) + (106{,}5803\ x\ 80)}{2060}$$

$$= \frac{218235{,}713}{2060} = 105{,}9397$$

Step 2: Calculate the Y-coordinate.

$$Y = \frac{(10{,}0706\ x\ 250) + (10{,}0151\ x\ 200) + \ldots + (11{,}3104\ x\ 90) + (10{,}9839\ x\ 80)}{2060}$$

$$= \frac{21239.941}{2060} = 10{,}3107$$

2.6. Results and Recommended General Area

2.6.1. Visualization of the Result

The Center of Gravity (COG) analysis was conducted using the geographical coordinates and weekly pallet and the calculation produced the following weighted coordinates: (10.3107,105.9397)

The Center of Gravity that is already calculated is shown up on Google Maps with other destinations. From the visualization, it illustrates that the COG is located nearby Vĩnh Long, Đồng Tháp và Tây Ninh. This means the optimal service area can be one of three provinces and became the the primary candidates for further evaluation. These provinces were selected due to their geographical proximity to the COG point and relatively high customer demand within Mekong KVD's distribution network. By applying some criteria related to logistics, the team will evaluate each potential destination to find the final best one.



Figure 2. Visualization of the Center of Gravity and Selected Candidate Areas

2.6.2. Recommended General Area

Criteria	**Vĩnh Long**	**Đồng Tháp**	**Tây Ninh**
Distance to COG	Closest	Near	Relatively Near
Centrality	Excellent	Good	Good
Access to customers	Excellent	Good	Good

Highway and Waterway Connectivity	Excellent	Good	Excellent
Industrial Park	High	Moderate	Moderate
Expansion Opportunities	High	Moderate	Moderate
Overall Suitability	Excellent	Good	Good

Table 3. Comparative analysis of three potential locations

According to the result of COG and comparative analysis, the team find out the best appropriate location for the introduction of Mekong KVD's Mega Hub is Vĩnh Long which given its proximity to the Center of Gravity (COG) point and the high level of customer demand in the area. To be more specific, Vinh Long currently hosts the Regional Distribution Center (DC) for Bach Hoa Xanh (in Binh Minh), as well as Co.opmart Vinh Long and GO! Ben Tre, making it the largest demand cluster within the entire network. Consequently, the company can promptly increase supply to meet this demand. Furthermore, Vinh Long's central location relative to other partners facilitates more flexible and convenient goods transportation.

Most importantly, Vinh Long boasts a well-developed transport infrastructure network, including National Highway 1A and proximity to both the Trung Luong – My Thuan Expressway and My Thuan Bridge, enabling the integration of road freight and inland waterway transport. Therefore, locating a "Mega Hub" in Vinh Long can help reduce transit times and lower transportation costs.

3. Choosing the Industrial Park

3.1. IP selection criteria

Selection of the suitable industrial park is one of the most important steps that is to be taken during the implementation of the Mekong KVD Mega Hub project as it has a direct effect on the financial feasibility and performance of the project in general. Besides, the selected location has to meet the following two conditions. First of all, the total amount of investments that include lease cost, construction costs, infrastructure development, and the construction of the cold chain system should not exceed 300 Billion VND. Secondly, the selected industrial park should have no less than 30 years left for the lease period. A sufficient lease period gives the

opportunity to distribute the costs of the investments over the period of time and recover the invested money, operate stably, and expand the business.

3.2. Comparing the three industrial zones

Source: (*Hoa Phu Industrial Zone - Vĩnh Long Province*, n.d.)

Source: (*Khu Công Nghiệp Bình Minh - Tỉnh Vĩnh Long*, n.d.)

Source: (*Long Hậu Industrial Park - Long An (Tây Ninh)*, n.d.)

Criteria	**Hoa Phu IP (Vinh Long)**	**Binh Minh IP (Vinh Long)**	**Long Hau IP (Tây Ninh)**
Land Lease Price	**160 USD/m²**	**100–120 USD/m²** (110 used)	**100 USD/m²**
Lease Term Remaining	Until **2058**	Until **2056**	More than 30 years remaining
Distance to Airport	25 km	15 km	25 km
Seaport Access	Can Tho Port (35 km)	Binh Minh Port (inside IP), Can Tho Port (16 km)	SPCT Port (3 km), Hiep Phuoc Port (3 km)
Water Supply	15,000 m³/day	7,000 m³/day	11,000 m³/day
Wastewater Capacity	6,900 m³/day	5,000 m³/day	8,000 m³/day

Main Industries	Food processing, pharmaceuticals, electronics	Food processing, garments, logistics	Logistics, food processing, supporting industries

Table 4. Comparison of Location, Infrastructure, and Investment Conditions of Candidate Industrial Parks

Capital Expenditure (Assumption Target Land = 30,000 m²)

Item	Hoa Phu	Binh Minh	Long Hau
Land Lease Cost **(Billion VND)**	**125.76**	**86.46**	**78.60**
Shell & Core Construction **(Billion VND)**	**90.00**	**90.00**	**90.00**
Estimated Total CapEx **(Billion VND)**	**215.76**	**176.46**	**168.60**
Company CapEx Budget Ceiling **(Billion VND)**	300	300	300
Available Budget Surplus / (Deficit) **(Billion VND)**	**+84.24**	**+123.54**	**+131.40**

Table 5. Capital Expenditure Comparison

3.3. Weighted Scoring Model

Scoring scale: 1 = Poor, 2 = Fair, 3 = Satisfactory, 4 = Good, 5 = Excellent.

Weighted score = Score × Weight.

Criteria	Weight	Hoa Phu IP	Binh Minh IP	Long Hau IP
Capital Expenditure (CapEx)	**35%**	**3** (1.05)	**4** (1.40)	**5** (1.75)
Logistics & Market Accessibility	**30%**	**4** (1.20)	**5** (1.50)	**3** (0.90)
Infrastructure & Utilities	**20%**	**5** (1.00)	**4** (0.80)	**5** (1.00)
Remaining Lease Period	**10%**	**5** (0.50)	**5** (0.50)	**5** (0.50)
Strategic Growth Potential	**5%**	**5** (0.25)	**4** (0.20)	**4** (0.20)

Total Weighted Score	**100%**	**4.00**	**4.40**	**4.35**
Rank	—	**3**	**1**	**2**

Table 6. Weighted Scoring Model

Three industrial parks are evaluated based on five key criteria: capital expenditure (CapEx), infrastructure and utility connection, geographic location and logistics connectivity, remaining lease period, and future expansion potential.

Hoa Phu Industrial Park is the most developed one of the three. It boasts reliable infrastructure such as a water supply of 15,000 m^3/day, a 6,900 m^3/day centralized waste water treatment facility and an uninterrupted power grid of 22 kV from the national grid, which makes this industrial park suitable for continuous production processes. Moreover, Hoa Phu Industrial Park is situated right next to National Highway 1A, being just 32 kilometers away from Can Tho City and 34 kilometers away from Cai Cui Port. It is projected that there will be Phase III extension of the industrial park by 157 ha. However, the cost of leasing land of Hoa Phu Industrial Park of USD 160/m^2 is higher than at other two industrial parks, making it the most expensive option for the investment with the highest initial costs. Phase I is already fully occupied and Phase II has leased over 80% of its industrial land.

The Binh Minh Industrial Park is the best option that gives a good balance between investment and operational effectiveness. The cost of leasing lands in Binh Minh Industrial Park is USD 100-120/m^2, which is much more economical than in the case of Hoa Phu. In terms of logistics, it has a strategically convenient location because it is located at a distance of 5 km from Can Tho City, 15 km from Can Tho International Airport, and 16 km from Can Tho Port. Binh Minh Port is located in the Binh Minh Industrial Park. It has a power plant of 110/22 kV, the water supply system with the capacity of 7,000 m^3/day and 5,000 m^3/day of wastewater treatment capacity. It provides good conditions for food manufacturing and cold chain operation. Most importantly, around 88% of B2B partners of the company are located in the Mekong Delta region such as Vĩnh Long, Cần Thơ, Đồng Tháp, An Giang, and Bến Tre.

Though Long Hau Industrial Park registers the lowest CapEx as well as providing good connectivity to Ho Chi Minh City and other international ports, it is not an ideal location for the firm because it favors firms that target the export market or have their customers based in Southeastern Vietnam. The primary reason for this is that the majority of the company's customers are in the Mekong Delta.

From the analysis above, Binh Minh Industrial Park has been chosen as the ideal site. Although Hoa Phu has better utilities and a well-developed industrial environment, the high price of the land makes the investment less efficient. On the other hand, while Long Hau has better facilities for exports, it is not as good in terms of customers' distribution. Therefore, Binh Minh Industrial Park has been identified to have the best combination of cost and other factors.

4. Delivery Strategy

4.1. Evaluation of Transportation Alternatives

Mekong KVD distributes FMCG products to 15 major B2B partners which are located across Western Ho Chi Minh city and Mekong Delta area. There are two transportation modes chosen for the Mega Hub: road freight and inland waterway transportation.

4.1.1. Transportation cost

Transportation cost is one of the most important factors in designing a logistics network because it affects the operating costs of a company as well as creating competitive advantages.

Road freight is the most popular mode of transportation for FMCG distribution because of its flexibility and direct delivery without carrying huge cargo containers. In addition, trucks have different sizes and flexible delivery schedules which allow companies to respond quickly to the customer demands. However, road freight has high transportation costs because it requires a lot of fuel, toll fees, maintenance, drivers' salary and other operating expenses. According to the World Bank, transportation makes up approximately 60% of total logistics costs in Vietnam (Lam et al., 2019), making transportation efficiency a key determinant of logistics performance.

Inland waterway is another suitable transportation mode especially across the Mekong Delta area. With barges, the company can carry heavy bulks for long distances. Compared with road freight, barges require lower transportation costs per unit, consume less fuel per ton of cargo. According to the Worldbank, one litre of fuel can transport one tonne of freight for

approximately 218 km by barge, compared with only 25 km by truck. This indicates that transport by barges is more efficient by 8.7 times than road transport (Amos et al., 2009).

For road freight more than 30–200 km, the average transportation cost is about from 2,500 to 2,700 VND per ton-km, while for 200–500 km, the cost remains relatively high at 2,050–2,200 VND per ton-km. In contrast, inland waterway transportation costs only 207–744 VND per ton-km over the same distance, showing a cost-saving advantage for transporting large volumes of cargo. (Lam et al., 2019)

Transport Distance	Unit	Road	Inland Waterways
<30 km	VND per ton-km	3,200–3,500	1,100–3,500
30–200 km	VND per ton-km	2,500–2,700	207–744
200–500 km	VND per ton-km	2,050–2,200	207–744

Table 7. Transportation Cost for Freight

The lower fuel consumption and larger carrying capacity of barges enable considerable economies of scale, particularly for palletized FMCG shipments destined for the Mekong Delta.

4.1.2. Delivery time

Delivery time is also an important factor when choosing the transportation mode because it affects directly to the customers' satisfaction and the process of filling inventory in the B2B distribution network

Road freight has a shorter delivery time because trucks can move directly from the Mega Hub (Central Hub) to the retail centers without arriving at intermediate cargo locations. This door-to-door delivery makes road freight become appropriate with the supermarkets and distribution centers that need frequent deliveries and filling processes.

However, inland waterway transportation requires longer times because barges travel at lower speeds than trucks and often require additional activities such as loading, unloading and transhipment before the goods reach the final destination. As a result, inland waterways are more suitable for planned shipments than urgent deliveries like road transportation.

For example, shipments from Cái Cui Port (Cần Thơ) to Cát Lái Port (Ho Chi Minh City) require approximately 4 hours by truck, compared with 24 hours by barge. Similarly, deliveries from Cái Cui to Cái Mép Port take around 6 hours by truck but approximately 36 hours by

barge. These differences demonstrate that road freight is a better choice for FMCG deliveries (Saigon Newport Corporation, 2021).

4.1.3. Reliability

Reliability refers to the ability to deliver products consistently according to planned schedules.

Road freight provides door-to-door delivery but it is affected by traffic congestion, road accidents, or breakage. According to the TomTom Traffic Index (2025), travelling only 10 km during evening peak hours in Ho Chi Minh City requires an average of 40 minutes and 32 seconds, with vehicles moving at an average speed of only 14.8 km/h. So, congestion can increase delivery delays, fuel consumption, and reduce trucks' life cycles.

Inland waterway transportation is unaffected by traffic congestion. However, delivery schedules may still be influenced by factors such as: seasonal water levels, port waitings, navigation restrictions, and weather conditions. (Calderón-Rivera et al., 2024). Furthermore, inland waterway transport is less flexible than road transport because it depends on fixed waterways, loading and transshipment operations, and scheduled services rather than flexible door-to-door delivery (OECD, 2006). Therefore, road freight is generally more reliable for time-sensitive FMCG deliveries, while inland waterways are more suitable for planned bulk shipments.

4.1.4. Capacity

Transportation capacity refers to the amount of cargo that can be transported in one trip and it directly affects shipment frequency and logistics costs.

Road freight is suitable for frequent FMCG deliveries but is constrained by vehicle size and legal weight regulations. Under Vietnam's road transport regulations, heavy truck combinations are generally limited to a legal gross vehicle weight of approximately 26–40 tonnes, depending on axle configuration.

In contrast, inland waterway transport offers substantially higher carrying capacity. According to PLC Corporation (n.d.), barges operating in Southern Vietnam are commonly available with capacities ranging from 200 to 500 tonnes, while larger barges can carry up to 1,500 tonnes, depending on route conditions and customer requirements. Therefore, a single barge can transport a huge number of cargo compared to road vehicles, allows for the consolidation of larger shipments and a reduction in the number of transport trips. This makes

inland waterways more suitable for high-volume FMCG distribution, whereas road freight is better suited to smaller and more frequent deliveries.

4.1.5. Environmental impact

Environmental sustainability has become an important concern in modern logistics management. Road freight creates high greenhouse gas emissions because it relies heavily on diesel vehicles. According to the World Bank (2024), transportation by trucks accounts for approximately 80% of greenhouse gas emissions generated by Vietnam's transport sector.

In contrast, inland waterway transportation is more environmentally sustainable. The World Bank (2019) showed that greenhouse gas emissions from inland waterway transport are two to three times lower than road freight. In addition, the World Bank (2024) also showed that road transport can emit up to six times more carbon dioxide than inland waterways. Because the use of fuel is more efficient therefore contributes to both lower operating costs and reduced environmental impacts.

4.2. Recommended Delivery Strategy

4.2.1. Recommended Transportation Mode

After selecting Mega Hub located at Binh Minh Industrial Park, road freight is recommended for distributing Mekong KVD's FMCG products to the 15 B2B customers in the Mekong Delta. Although inland waterway provides lower transportation costs and higher capacity for large bulk transportation, road freight provides better flexibility, faster delivery, and direct door-to-door service to warehouses of 15 partner locations.

Binh Minh industrial park is located near National Highway 1A and Can Tho City, allowing convenient access to provinces in the Mekong Delta. Mekong KVD with products such as fish sauce, soy sauce, and cooking oil that require frequent replenishment to retailers and distributors, road transportation helps reduce lead times and improve delivery reliability. These fragile products are not appropriate for many processes for handling. Therefore, road freight is the most appropriate transportation mode to support the company's distribution strategy and customer service objectives.



Source: batdongsan.com

Figure 3. Perspective view of Binh Minh Industrial Park.

4.2.2. Distribution Network

The new distribution network adopts a direct distribution model. Finished products are transported from the manufacturing centers to the Mega Hub at Binh Minh Industrial Park before being delivered directly to the 15 B2B customers across the Mekong Delta.

After receiving orders from manufacturing centers, products are stored, and managed in the warehouse. When partners place orders, products are then picked from inventory, consolidated based on delivery routes, palletized, and loaded to trucks for distribution. Delivery routes should be planned in advance to serve customers in the same geographical area during one trip. By consolidating multiple customer orders into larger shipments, the Mega Hub takes advantage of the vehicles storage space, reduces transportation costs, and minimizes unnecessary handling which can help decrease damaged glass bottles.

4.2.3. Expected Benefits

The recommended transportation strategy is expected to improve logistics performance and customer service. By consolidating multiple orders at the Mega Hub helps optimize vehicle capacity and reduce transportation costs. Road freight transhipment allows shorter lead times, increased reliability and flexibility. In addition it also reduces the ability of products breakage and damage during transporting. Overall, this strategy not only ensures the effective distribution but also enhances customer satisfaction, while creating a foundation for a scalable logistics network for future growth of the business.

5. Material Handling

5.1. Current Problem

Mekong KVD distributes FMCG such as fish sauce and soy sauce in glass bottles, which are highly fragile when packaging, transporting and material handling. Glass packaging is more vulnerable to vibration than plastic packaging; consequently, it is easier to damage if not properly protected (Yam, 2009). The fragmented logistics network requires products loaded, unloaded, and transported many times between different facilities. That increases the risk of bottle breakage. Furthermore, incorrect use of pallet, forklift, and vibrations during transport can make the cargos more likely to experience damage. These issues not only raise logistics costs, reduce operational efficiency but also customer satisfaction.

5.2. Proposed Improvements

To address the current high breakage rate of glass products, an appropriate packaging and material handling system is needed to ensure product stability, minimize impact of external factors, and protect the products throughout the transportation and distribution process.

5.2.1. Packaging Improvements

Step 1. Pack glass bottles in heavy-duty corrugated cartons with dividers

Each glass bottle is packed in a double-wall carton (5-ply) with corrugated cardboard dividers separating individual bottles. The dividers prevent collision between bottles, while the thick carton layers absorb shocks and compression when transporting. This significantly reduces the risk of breakage caused by vibration and impact.



Figure 4. Double Wall Sheets

Step 2. Stack cartons onto standard pallets

After putting in the cartons, cartons are stacked on standard pallets using a column stacking pattern, where each carton is placed directly above the one beneath it. This arrangement allows the weight to be transferred vertically through the strongest parts of the cartons, which is four angles, maximizing their compression strength and reducing the problem of carton deformation.



Figure 5. Column stacking pattern

Step 3. Place anti-slip foam sheets between carton layers

To enhance the stability, anti-slip foam sheets are put between each carton. These sheets increase friction between the cartons, preventing them from sliding when transporting. This

helps maintain the integrity of the pallet and reduce the risk of carton movement or bottle breakage.





Figure 6. Anti Slip Layer Sheets

Step 4. Secure the pallet with polyester strapping

The entire pallet load is reinforced using polyester (PET) strapping. The straps tightly hold the cartons together, helping the entire pallets avoid shifting or collapsing during handling and transport. PET strapping is used because it provides high tensile strength while remaining flexible enough to absorb vibrations.



Figure 7. Polyester Strapping

Step 5. Install corner protectors

Plastic or cardboard corner protectors are used to cover all four vertical corners of the pallet before strapping. These protectors distribute the tension from the straps evenly, preventing damage to carton edges and improving the structural stability of the palletized load.





Figure 8. Corner Protectors

Step 6. Wrap the pallet with stretch film

After strapping, the entire pallet is wrapped with multiple layers of stretch film (PE stretch wrap). The stretch film secures all cartons into one stable unit, protects the products from dust and moisture, and further prevents cartons from shifting during transportation.





Figure 9. Stretch Film PE

Step 7. Load pallets using forklifts

An electric forklift is used to lift finished pallets onto the truck. Handling using specialized mechanics helps minimize manual lifting and reduces the risk of dropping or impacting fragile products. It also improves efficiency, loading speed and workplace safety.



Figure 10. Forklift lifting pallets on truck

Step 8. Secure pallets inside the truck

Inside the truck, pallets are protected by using cargo straps or load bars. These restraints prevent pallets from moving or sliding during braking, or traveling to poor infrastructure roads. Keeping the pallets in place and decreasing breakage during transit.



Figure 11. Cargo Strap



Figure 12. Load bar

5.3. Expected Benefits

These improvements will help reduce glass bottle breakage by ensuring the stability and protection of the fragile FMCG products during transportation. They will also lower logistics costs by reducing damaged products and making loading and unloading faster. In addition, using pallets and forklifts improves work efficiency. As a result, customers will receive their orders safely and on time, leading to higher customer satisfaction.

6. Conclusion

The movement to a single, integrated Mega Hub is a strategic decision to overcome operational inefficiencies, reduce logistics costs and minimise glass bottle breakage, and follows the consolidation of Mekong KVD's four fragmented facilities. The Center of Gravity (COG) analysis was performed according to the spatial distribution and weekly demand of 15 B2B partners with total demand of 2,060 pallets and revealed that the optimal weighted location is at coordinates $(10.3107, 105.9397), indicating Vĩnh Long as the best general area due to its central position relative to high demand clusters.

A weighted scoring model was applied to evaluate three industrial parks to determine the exact location for the facility with a 300 Billion VND CapEx ceiling and a minimum 30-year lease requirement. Bình Minh Industrial Park (Vĩnh Long) was the choice with the highest total score of 4.40. It hits the right mix of financial efficiency, with an estimated CapEx of 176.46 Billion VND (with a surplus of 123.54 Billion VND), and strategic utility, with a long lease term until 2056, dedicated port infrastructure and direct access to National Highway 1A and Can Tho City.

Inland waterways were rejected in favour of road freight with a direct distribution model for the delivery network. Waterways are cheaper per unit, but road transport offers the speed, door-to-door flexibility and reduced cargo handling that sensitive FMCG goods require. In addition, a standardized 8-step packaging and material handling system was installed using 5-ply cartons with dividers, anti-slip foam, PET strapping and load bars inside the trucks to avoid product damage in transit.

In summary, the combination of the Bình Minh Mega Hub location, direct road transportation and stringent material handling standards provides Mekong KVD with a

scalable, cost-efficient, low-breakage supply chain structure that ensures high service quality for the Mekong Delta area.

7. Executive summary & strategic D2C context

7.1. Strategic Transition from B2B to D2C

After researching and analyzing in the previous part, a new Mega Hub of Mekong KVD will be located at the Binh Minh Industrial Park (Vinh Long) to optimize logistics costs, ensure meeting a high customer demand promptly and minimize product damage during regional distribution. Although this strategy can enhance the efficiency of the wholesale distribution channel, Vietnam's fast-moving consumer goods (FMCG) market is shifting rapidly toward digital commerce. Due to the rising demand for consumer goods, customers expect manufacturers to offer direct online purchasing options alongside traditional retail outlets. Therefore, only heavily relying on supermarket partners will restrict the Mekong KVD's ability to generate higher profit margins and build direct relationships with end consumers.

From there, to improve its competitiveness, the Board of Directors decided to expand into a direct-to-consumer (D2C) business model via a dedicated e-commerce platform. Different from the B2C model, D2C approaches allow Mekong KVD to interact directly with customers, gather first-party consumer data, personalize promotions, and strengthen long-term brand loyalty.

However, it is more difficult for Mekong KVD when shifting a B2B to a D2C. Because it can affect its logistic management. To be more specific, the company has to control thousands of individual orders instead of delivering full pallets to distribution centers. And each entails distinct processes such as picking, protective packaging, and last-mile delivery. Consequently, the successful probability of a D2C initiative depends not only on building an online sales platform but also on designing an order fulfillment strategy that balances customer service, delivery speed, logistics costs, and profitability.

7.2. Operational Bottlenecks in Liquid & Glass E-Commerce Logistics

Although the D2C model can bring more growth opportunities for Mekong KVD, there are also some operational challenges for Mekong KVD’s product portfolio, including Fish sauce, soy sauce, and cooking oil. They are all liquids primarily packaged in glass bottles , leading to make the order fulfillment process both costly and operationally complex.

The first challenge is the heavy freight of goods. This is because a standard product bundle consists of three bottles with a total weight of approximately 3 to 5 kg, depending on the bottle 1 size. As a result, shipping costs may be higher, and delivery productivity can be reduced because drivers can carry fewer orders per trip.

The second challenge concerns fragile glass packaging. Glass bottles are easier to crack, leak which lead to damage the quality of products during sorting, transportation, and delivery. Moreover, the pallets will go through multiple transit stages further which increases the risk of product damage before it reaches the customer. Therefore, damaged orders negatively impact customer satisfaction, drive up return rates, and risk harming the brand's reputation. In order to solve this problem, the company is considering the use of specialized protective packaging such as air-column bags or air-cushion wraps is essential, even though this adds approximately 25,000 VND to the cost per order based on the project's assumptions.

Finally, the D2C model can result in the threat to profit margins on individual orders. The reason for that concern is each order incurs relatively fixed processing costs, including picking, protective packaging, and last-mile delivery. For low-value, individual orders, these operating costs represent an excessive proportion of revenue and can even completely wipe out profits.

7.3. Geographic Scope for Phase 1 Pilot Launch

Instead of launching nationwide immediately, this report suggests selecting Can Tho City as the pilot market for Phase 1. The first reason for this decision is Can Tho serves as the important economic and logistics hub of the Mekong Delta. At the same time, Can Tho City has the highest urban population densities and a relatively well-developed transport infrastructure (ACC Hanoi, 2025). Additionally, this city is one of Mekong KVD’s two largest B2B clients, including GO! Can Tho (200 pallets/week) and MM Mega Market Hung Loi (180 pallets/week) which account for a significant share of pallet distribution in the area. By leveraging these existing retail partners allows Mekong KVD to pilot an integrated logistics model combining wholesale and online sales operations without requiring substantial investment in additional infrastructure.

Most importantly, according to the previous part in 3.3 and 4.2 of Preliminary #1, the Mega Hub located at Binh Minh Industrial Park (Vinh Long) is situated in close proximity to National Highway 1A and Can Tho City. To be more specific, It is 5 km from Can Tho City, 15 km from Can Tho International Airport, and 16 km from Can Tho Port. With shorter delivery

distances, the company can reduce transportation costs while enabling same-day or even two hour delivery services with the support of local carriers. Furthermore, focusing on putting more 2 inventory at existing supermarkets also makes Can Tho an ideal environment for evaluating micro-fulfillment strategies before scaling up nationwide.

Finally, locating and managing only the small scale of the pilot to a single urban market helps company to track customer purchasing behavior, evaluate order processing efficiency, and identify operational bottlenecks before committing to a major investment for nationwide rollout.

8. Comparative evaluation of D2C fulfillment models

After completing the construction of Mega Hub, Mekong KVD needed to determine the most suitable order fulfillment strategy for its new Direct-to-Consumer (D2C) platform. Based on their core products that are heavy liquid products packaged in fragile glass bottles, the fulfillment model had to meet requirements such as balanced logistics costs, delivery speed, customer experience, inventory efficiency, and long-term scalability. Therefore, this section aims to compare two viable options: a standalone Multichannel strategy and an integrated Omnichannel strategy.

8.1. Model A: Multichannel Logistics Strategy

As part of its multichannel strategy, Mekong KVD's D2C business operates independently of its existing B2B network. Specifically, orders placed via the website or mobile app are processed and dispatched directly from the Mega Hub or a dedicated "dark store" located in the center of Can Tho. When products are confirmed and already picked and packed using specialized protective materials such as air-column bags . Then they are being delivered to customers by third-party logistics (3PL) providers like GHN, Viettel Post or J&T Express. Inventory for the online sales channel is managed separately from B2B stock, ensuring complete operational independence.

The remarkable advantage of this model is the ability to control and maintain absolute control over the customer experience and purchasing data. In addition, when choosing this model, the company can retain 100% of retail revenue without paying commissions to retail partners, allowing for greater flexibility in pricing and the implementation of promotional campaigns. Mekong KVD may supervise and manage the whole the order fulfillment process with each specific stage, including order processing and protective packaging to after-sales

service, thereby ensuring consistent quality standards and fostering stronger customer relationships.

However, the multichannel model will increase the logistic costs. This is because under this model, the last-mile delivery costs 80,000 VND per order, excluding an additional 25,000 VND fee for specialized shock-resistant or breakage-proof packaging. Furthermore, estimated delivery times typically range from one to two days, which is slower than hyper-local express delivery models. Besides, the company has to consider the establishment or rental of a dedicated warehouse (or "dark store") carefully, because it requires significant upfront investment, covering warehouse rental, staffing, a warehouse management system (WMS) and daily operating expenses.

Overall, the multichannel strategy will help the company to tight operational control and ownership of customer data, but it also demands higher logistics and infrastructure costs; this can erode profitability for low-value fast-moving consumer goods (FMCG) orders.

8.2. Model B: Omnichannel Fulfillment Strategy

In terms of omnichannel strategy, Mekong NVD will integrate its D2C platform with a network of 15 existing B2B retail partners. Thus, some retailers such as Co.opmart, GO!, MM Mega Market and Bách Hóa Xanh serve as micro-fulfillment centers (MFCs). To be more specific, a Distributed Order Management (DOM) system automatically routes customer orders to the nearest store with available stock. After that, orders are delivered through several local logistics partners like GrabExpress, Ahamove or customers can choose to pick up their purchases themselves by using the "Buy Online, Pick Up In-Store" (BOPIS) model.

The key advantage of this model is the speed of delivery and ordering process. Under the omnichannel strategy, the time delivery is faster and more cost-effective than the order fulfillment process. By leveraging nearby retail locations, last-mile delivery costs drop from 80,000 VND to 35,000 VND per order, while delivery times are reduced to just 1–2 hours. This shared inventory network also improves inventory turnover and minimizes excess stock across both B2B and D2C channels.

In summary, the omnichannel model delivers faster delivery speeds, lower logistics costs, and improved inventory efficiency, making it a powerful solution to support Mekong KVD’s D2C expansion strategy.

8.3. Comparative Summary of the Two Fulfillment Models

Criteria	Multichannel	Omnichannel
Fulfillment location	Mega Hub / Dedicated Dark Store	Existing retail partners (Micro-Fulfillment Centers)
Delivery cost	80,000 VND/order	35,000 VND/order
Delivery time	1-2 days	1–2 hours
Retail commission	None	25% of gross sales
Inventory	Separate B2B and D2C inventory	Shared inventory across B2B and D2C
Customer experience	Consistent but slower	Faster and more convenient
Technology requirement	Basic WMS and e-commerce platform	DOM system with real-time inventory synchronization
Initial investment	Higher warehouse investment	Higher software integration investment
Scalability	Limited by warehouse capacity	Highly scalable through existing retail network

Table 8. Comparison of Multichannel and Omnichannel Fulfillment Strategies

Based on the table of comparison between two models, the integrated omnichannel strategy demonstrates superior performance in terms of delivery speed, logistics efficiency, and inventory utilization; meanwhile, the multichannel strategy offers greater operational control and eliminates retail commission costs. The following section provides a quantitative assessment of both approaches to determine which strategy yields the highest financial performance for Mekong KVD.

9. Mathematical and financial modeling

9.1. Baseline Product Economics & Gross Margin

Before evaluating the feasibility of the D2C model, Mekong KVD should establish a baseline standard product offering. The company offers a “Mix and Match” bundle including three bottles of fish sauce, soy sauce and cooking oil. This allows customers to select a combination of bundles.

Items	Value (VND)
Retail selling price (3 bottles)	300,00
Cost of goods sold (COGS)	120,00
Gross profit	180,00
Gross margin	60,00%

Table 9. Baseline Product Economics

From the table, each bottle costs VND 100,000 leading to the total of the bundle of VND 300,00. The cost of goods sold (COGS) is calculated at VND 40,000 for each bottle, which means VND 120,000 per bundle. Therefore, the bundle generates a gross profit of VND 180,000 showing that the gross margin is about 60% before adding any of the logistics expenses.

9.2. Unit Economics Comparison Matrix

To evaluate the financial performance between the two D2C models which are multichannels and omnichannels, an analysis was conducted based on one customer order. This analysis compares the primary operating and logistics costs related to each model including delivery, retail commission, and protective packaging.

Cost Item	Multichannel (VND)	Omnichannel (VND)
Retail revenue	300,000	300,000
Cost of goods sold (COGS)	120,000	120,000
Gross profit	**180,000**	**180,000**
Retail commission		75,000
Last-mile delivery	80,000	35,000
Anti-breakage packaging	25,000	25,000
Net contribution margin	**75,000**	**45,000**
Net contribution margin (%)	**25%**	**15%**

Table 10. Unit Economics Comparison per Order

As shown in the table, the Multichannel model generates a higher net contribution margin at VND 75,000 per order (25%). It is because the company does not pay commission to retail partners. However, the model requires a higher last-mile delivery cost of VND 80,000 per order because of direct delivery from the Mega Hub or a dedicated Dark Store.

In contrast, the Omnichannel model with lower delivery cost of VND 35,000 per order by utilizing existing retail partners as micro-fulfillment centers. Nevertheless, the company is 6 required to pay a 25% retail commission (VND 75,000 per order), resulting in a lower net contribution margin of VND 45,000 (15%).

Overall, Multichannel provides a higher profit per order, Omnichannel lowers last-mile delivery costs by leveraging the retail network.

9.3. Packaging Cost Allocation Strategy (Anti-Breakage Fee)

KVD products with glass bottles require protective packaging to reduce the risk of breakage during the delivery. In this study, the anti-breakage packaging cost is calculated at VND 25,000 per order. Therefore, Mekong KVD must decide whether to absorb this cost or charge it to customers as a handling fee. There are two scenarios that will be evaluated.

In the first scenario, the company pays full packaging cost. As a result, the net contribution margins remain 25% for the Multichannel model and 15% for the Omnichannel model, as shown in Table 10. Although this approach reduces the company's profit per order, customers benefit from a transparent pricing policy with no additional charges. This strategy is also expected to improve purchase intention and reduce cart abandonment.

In the second scenario, customers pay approximately VND 15,000–25,000 per order for the handling fee. This approach enables the net contribution margin to increase by recovering part or all of the packaging cost. However, this additional fee may discourage customers from purchasing their products, particularly for orders with low value, which leads to a higher cart abandonment rate.

Because of the customer experience during the initial stage of D2C implementation, this study recommends that Mekong KVD should pay full cost of anti-breakage packaging for orders that meet the Minimum Order Value (MOV). This approach allows customers not only to benefit from free packaging costs which makes the purchasing process more attractive but also encourages larger order values. The importance of MOV in maintaining profitability is discussed in the next section.

9.4. Minimum Order Value (MOV) & Sensitivity Analysis

The Minimum Order Value (MOV) analysis was considered to determine the minimum purchase needed for Mekong KVD to achieve a net contribution margin at a minimum percentage of 20% under the Omnichannel model. As delivery and anti-breakage packaging 7 are largely fixed costs per order, increasing the number of bundles allows these costs to be divided across a larger sales value, resulting in improving profitability.

The net contribution margin is calculated using the following equation:

$$\text{Net Margin} = \frac{(NxRSP) - (N\ x\ COGS) - Commission - Delivery\ Cost - Packaging\ Costs}{N\ x\ RSP}$$

Where:

N = Number of bundles per order

RSP = Retail Selling Price per bundle

COGS = Cost of Goods Sold per bundle

After applying the formula, the results are shown in Table 11

Number of Bundles	Revenue (VND)	Net Contribution	Net Margin
1	300,000	45,000	**15%**
2	600,000	150,000	**25,00%**
3	900,000	255,000	**28,33%**

Table 11. Sensitivity Analysis of Minimum Order Value (Omnichannel).

The analysis shows that a single bundle generates a net contribution margin of only 15%, which is below the company's target (20%). However, when customers purchase two bundles, the margin increases to 25% which is higher than the target because fixed fulfillment costs are divided into a higher order value. The profitability continues to improve when customers place larger orders. Setting the MOV at two bundles (VND 600,000) provides a balance between customer affordability and financial performance.

Therefore, Mekong KVD should establish a Minimum Order Value of VND 600,000 (two bundles). Orders meeting this level should receive free delivery and free anti-breakage packaging. Meanwhile, orders below the MOV may be subject to delivery or handling charges to partially recover fulfillment costs.

9.5. Upfront Capital Investment and Technology Analysis

To achieve high profitability per order, the initial investment should also be considered when selecting a suitable D2C logistics strategy. Although the Multichannel model generates a higher net contribution margin per order, it requires larger upfront investment to establish an independent fulfillment network.

With the Multichannel model, Mekong KVD needs to rent or operate a Dark Store, purchase warehouse equipment, and employ workers to process online orders. These requirements

increase capital expenditure (CapEx) and operating expenditure (OpEx).

In contrast, the Omnichannel model takes advantage of the existing infrastructure of 15 retail partners as micro-fulfillment centers. As a result, instead of building additional warehouses for the D2C channel, the primary investment is the Distributed Order Management (DOM) system. This system synchronizes inventory information and automatically routes customer orders to the most appropriate location. As a result, Mekong KVD can leverage existing resources effectively to build a completely new logistics network.

Investment Item	Multichannel	Multichannel
Dedicated Dark Store	✓	_
Warehouse equipment	✓	_
Warehouse staff	✓	_
Distributed Order Management (DOM) system	_	✓
System integration and employee training	✓	✓
Initial infrastructure investment	High	Low

Table 12. Comparison of Upfront Investment Requirements

This investment structure is aligned with the findings of Hübner et al. (2016), who reported that retailers transforming from multichannel to omnichannel integrate inventory and warehouse operations across sales channels rather than maintaining separate facilities. This integration offers more flexible and efficient inventory allocation. It also improves warehouses using space, and reduces the need for additional logistics infrastructure.

Implementing a DOM system may require initial technology investment as previous studies indicate that order management systems make inventory more visible, automate order routing, reduce fulfillment costs, and shorten order processing time. That leads to positive long-term financial returns. Forrester's Total Economic Impact (TEI) studies also suggest that organizations investing in advanced order management often achieve favorable returns on investment and short payback periods. It is because of savings on operational activities and order processing.

Therefore, the Omnichannel model generates a lower contribution margin per order than the Multichannel model, it offers lower infrastructure investment, utilizes existing resources more effectively, and offers greater scalability for future expansion. Considering both operational efficiency and capital requirements, the Omnichannel strategy represents a more financially sustainable solution for Mekong KVD's D2C implementation.

9.6. Final Strategic Selection Justification

Based on the analyses operationally and financially in the previous sections, the Omnichannel strategy is the most appropriate D2C model for Mekong KVD. The model leverages the company's existing network, reduces the need for additional infrastructure, and provides a scalable fulfillment system to support future expansion. In addition, the packaging cost strategy of Mekong KVD absorbs the anti-breakage packaging cost for orders meeting the Minimum Order Value (MOV). That helps maintain a competitive pricing while encouraging customers to place more orders. Together, these advantages strengthen both operational efficiency and customer experience during the D2C implementation.

10. D2C Operational & Last-Mile Execution Plan

10.1. Packaging Standards for Glass Bottle Protection

Because of the use of glass bottles in the packaging of Mekong KVD’s products, the need for safeguarding them during the last-mile delivery process becomes very important to avoid product damage and customer dissatisfaction. The difference between B2B pallet delivery and D2C delivery is that D2C order involves more handling steps such as picking, packing, sorting, and delivery to the customer premises. Thus, all orders made online should adopt the four-layer packaging process.

The packaging process commences by placing a tamper-proof safety seal on each of the bottle caps to ensure no leakages occur while the bottles are being transported. The bottles are wrapped in an air-column protective sleeve to cushion the bottles against any shocks that may lead to breakage of the bottles. Corrugated cardboard partitions are also placed between the bottles to avoid contact while being transported, and the bottles are placed in a branded D2C carton and secured with security tape and marked “Fragile – Glass Bottles”.

Based on the assumptions of the project, the total cost of the antibraking packaging system is VND 25,000 per order. In line with the recommendations in Section 3, Mekong KVD should

take the cost into account for all orders worth VND 600,000 as the Minimum Order Value (MOV).

Packaging Layer	Purpose
Tamper-evident safety seal	Prevent liquid leakage
Air-column protective sleeve	Absorb shocks and reduce bottle breakage
Corrugated cardboard dividers	Prevent bottle collision
Branded D2C carton with security seal	Protect products and reinforce brand image

Table 13. Four layer Packaging Standard

10.2. Distributed Order Management (DOM) System and Partner Workflow

For implementation of the Omni-channel approach, Mekong KVD plans to launch a Distributed Order Management (DOM) solution which will connect the D2C platform of the company to the POS systems of the participating retail stores like GO! Can Tho and MM Mega Market Hung Loi as a part of Phase 1 pilot. The use of DOM system ensures real-time tracking of inventory and helps allocate each order to the closest store with the relevant inventory automatically.

The fulfillment process starts when a customer makes an order via the website or mobile application of the Mekong KVD company. Once the payment is confirmed, the DOM system analyses the stock availability of participating supermarkets in real-time and decides on the appropriate fulfilment location. Next, the store personnel carry out necessary pick and pack operations following the standard packaging process of Mekong KVD. Finally, the package is picked up for delivery by GrabExpress or Ahamove, while customers can opt for the BOPIS option as well. The use of retail stores as the place of order fulfilment allows the company to combine online and offline inventories, increasing flexibility and efficiency of order fulfilment (Hübner et al., 2016). In addition, DOM systems provide real-time inventory visibility and automatically assign customer orders to the most suitable place of order fulfilment (Melacini et al., 2018).

The limitation of Phase 1 implementation to Can Tho City will allow Mekong KVD to measure its performance in a controlled setting and take advantage of its current sales network. Performance measurement can be done using factors such as order fulfillment period, inventory accuracy, punctuality, order accuracy, and product breakage percentage prior to implementing the omnichannel network in other provinces of the Mekong Delta.



Figure 13. Omnichannel Order Fulfilment Workflow

10.3. Risk Mitigation Framework

In order to guarantee that the operations run smoothly during Phase 1 piloting, it would be wise for Mekong KVD to consider addressing the two risks highlighted in the Omnichannel Fulfilment process, which are product breakage during the last-mile delivery and inventory mismatch in partner stores.

The first risk to be considered is product breakage that occurs when the orders are sent to GrabExpress or Ahamove. Even though the four-layer packaging will minimize this risk, damage can still be incurred during delivery. Consequently, Mekong KVD must consider having a two-hour breakage replacement policy, where customers complain about their broken orders using the website or mobile application and the DOM system generates a replacement order for the nearest partner store.

The second risk is inventory mismatch in partner stores as compared to the D2C platform. To limit stock shortages, each supermarket participating in the pilot will keep a safety stock buffer of five bundles per day, according to project assumptions. The DOM system will keep the inventory information up-to-date, whereas the Mega Hub will replenish stocks whenever

the inventory is below the safety stock level.

11. Conclusion & Integrated Logistics Roadmap

11.1. Synthesis of the Omnichannel D2C Strategy

The strategic transition to a Direct-to-Consumer (D2C) model is essential for Mekong KVD to capture digital market share and build long term brand loyalty. Following a rigorous financial and operational evaluation, the integrated Omnichannel fulfillment strategy emerged as the optimal solution over the standalone Multichannel approach. By leveraging the existing network of 15 retail partners instead of investing heavily in dedicated dark stores, Mekong KVD can significantly minimize initial capital expenditure.This model solves well the logistics problem of transporting large, fragile glass bottles, by minimizing the last-mile delivery cost to VND 35,000 and ensuring fast delivery in 1-2 hours. With a strategic Minimum Order Value (MOV) of 600,000 VND (two bundles), the Omnichannel model effectively absorbs the anti breakage packaging costs and keeps a sustainable net contribution margin at 25%.

11.2. Strategic Fit with the Mega-Hub

The success of this D2C initiative will rely on a seamless logistics network, which will be supported by the new Mega Hub at Binh Minh Industrial Park. In the integrated architecture the Mega Hub is the core of the wholesale B2B supply chain. The pallets are stored in bulk and then efficiently distributed to the local supermarkets. Existing retail partners such as GO! and MM Mega Market are being transformed into localized micro-fulfillment centers (MFCs) for the D2C channel. These are then synchronized via a Distributed Order Management (DOM) system to allow B2B replenishment to fuel D2C last mile execution, increasing inventory turns and eliminating excess warehousing.

11.3. Expansion plan

Mekong KVD will implement a structured 6-month pilot phase to manage operational risk, focusing mainly on the Can Tho City market, due to the short distance (5 km) from the Binh Minh Mega Hub.In this pilot phase, we will test the synchronization of the DOM system, optimize the specialized air column packaging processes and check customer acceptance of the MOV strategy. The company will realize stable unit economics and fulfillment metrics in this 6-month testing window and then begin a phased rollout, smoothly scaling the Omnichannel D2C model across the larger Mekong Delta region.

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